



MAMDOUH ATTIA

Software Engineer

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[Portfolio Link](#)

PROFESSIONAL SUMMARY

Computer Engineering graduate (Cairo University) with **3+ years** building **cross-platform** mobile apps for real clients, Academic projects, open-source projects, and AI-powered software. Delivered end-to-end solutions using **Flutter, Python, and C/C++**. Strong foundation in systems programming, **data structures, algorithms**, and **AI**. **200+** hours technical instruction across **ML, Python, Embedded**, and **Flutter** Programs.

EDUCATION

Cairo University — Faculty of Engineering, Computer Engineering

2019 – 2024

Grade: Very Good with Honors | Graduation Project Grade: Excellent (A+)

PROFESSIONAL EXPERIENCE

Flutter Developer — Freelance, Primalab, Itqan Community

Sep 2023 – Present

- Contributed to open-source package (mushaf-imad-flutter) ([PR #27](#)); Integrating Quran.com's audio API : OAuth 2.0, hybrid audio engine, Stream Transformers, 80+ Mocktail tests, 86% latency reduction (by token caching).
- Owned full SDLC Delivering +3 client Flutter apps (HyperMate, VolunTeen, Check-It) in MVC architecture & provdier state management.
- Integrated Hardware IoT sensors (ESP8266), AI chatbots (GPT-3), Firebase, REST APIs, QR & barcode scanning.

Technical Instructor — DEPI/MCIT @ Harvest | DECI/MCIT Eyouth | Maspero | Mindset | Primalab

Aug 2023 – Present

- DEPI Initiative:** Led 100+ hrs of ML Engineering for 15+ engineers (Maths - Python - ML - Azure AI); designed & graded 17 tasks
- Maspero Academy:** Delivered 20+ hrs Python basics for 35 students.
- Primalab:** 50+ hrs Dart/Flutter across 28 sessions for 10+ students. | **Mindset:** Taught System programming – Embedded systems – AI.

IT Support Specialist — Future Of Egypt (Mandatory Military Service)

Dec 2024 – Nov 2025

- Technical support for hardware, software, and networks in a government operational environment.

Flutter Developer Intern — Orange Digital Center Software House

Jun 2023 – Sep 2023

- Worked in Agile squads on 2 internal mobile apps and 1 hackathon e-wallet. Translated Figma designs to Flutter UI using MVVM + Bloc/Cubit.

SELECTED PROJECTS

[Mushaf-IMAD Package](#) | Dart, REST API, OAuth 2.0, Streams, Mocktail, GitHub, Open Source Contribution

Feb – Mar 2026

- 86% latency reduction via Single-Trip token optimization; hybrid timing hierarchy (RAM → Local → API) near 0 latency audio playback sync.
- +70 atomic commits of 5000+ lines, 80+ async Mocktail tests; published architectural guides for open-source maintainability.

[Predicto](#) (Graduation Project, A+) | Python, PyTorch, OpenCV, Desktop GUI

Aug 2023 – Jul 2024

- Team lead:** built vPredicto open-source library for Video Frame Prediction with full PyQt desktop app (model selection, video input, output).
- Implemented from scratch:** *SimVP* [co-implemented] (CNN Encoder-Translator-Decoder, TPU-optimized, 5× training speedup via checkpointing) and PredNet (ConvLSTM predictive coding, resolved critical dimension-mismatch bugs).

[VolunTeen](#) | Flutter, Firebase, GPT-3, Provider, MVC

Sep 2023 – Jan 2024

- Solo developer for real client: 30–40 responsive screens, GPT-3 chatbot, Firebase storage, in-app video player, WhatsApp/URL launcher.

[HyperMate](#) | Flutter, C, ESP8266, Wi-Fi, GPT-3

Nov – Dec 2023

- IoT stress-monitoring app with real-time GSR sensor readings over Wi-Fi, personalized exercise, AI chatbot. Also authored Arduino firmware.

[Distributed File System \(Go-DFS\)](#) | Go, gRPC, Protocol Buffers, TCP

Mar – Apr 2024

- Master-Worker architecture: gRPC heartbeat monitoring, auto-replication engine (factor 3), Data Keeper self-scanning, and MP4 upload/download pipelines via raw TCP.

[Gamingo Engine + Drive Me Home Game](#) | C++, OpenGL, GLSL, OOP

Mar – May 2023

- Custom 3D game engine: Role: Shader, Texture, Material, Sampler classes + GLSL programmable shading. Collision detection, game-state machine, post-processing shaders (Sepia, CRT, Inversion).

[MIPS Processor](#) | Verilog, Computer Architecture, FPGA Simulation, Modelsim

Dec 2022

- 5-stage pipelined MIPS processor. **Role:** Decode stage, top-level integration, control unit ISA, hazard detection forwarding verification.

[OS Scheduler](#) | C, Linux, IPC, Memory Management

Apr 2022

- Led Process Generator: fork()/exec() lifecycle, IPC Message Queues. Designed Buddy Allocator (1024-byte pool, recursive splitting/merging, allocate() + deallocate() functions).

SKILLS

Programming: Dart · C/C++ · Python · Java · Go · SQL | **Web (basic):** HTML · CSS · JS · Node.js · Jest · Database · Accessibility

Mobile / Frontend: Flutter · Provider · Bloc/Cubit · MVC · MVVM · Clean Architecture · Streams · BAAS (Firebase · Supabase) · REST APIs

AI / ML: PyTorch · TensorFlow · scikit-learn · OpenCV · Computer Vision · NLP · pandas · numpy · Machine Intelligence · ML · DL

Systems, CS & SWE: Linux · Git/GitHub · GitHub Actions CI/CD · Database · MySQL · Algorithms · Data Structures · Design Patterns · OOP · SOLID SDLC · Agile/SCRUM · gRPC/protobufs · Bash & shell scripting · Mockito · JSON

Hardware: ESP8266 · Wi-Fi · Arduino Firmware · Hardware Sensors · Embedded Systems · IC Design · Verilog · VLSI

CERTIFICATES

[GitHub Professional Certificate](#) (2026)

[Flutter Internship — ODC](#) (2023)

[Machine Learning Specialization — Coursera](#) (2024)

[HCIA-AI — Huawei International](#) (2021)